

Lab - DNSSec

Version 1.1, 18-11-2020, by Ivo Nijhuis

Introduction

In this lab we'll see how we can verify the validity of DNS records using DNSSec.

To participate in this lab you'll need Virtual Box and the NE1 Debian VM. You probably already have these set up from the previous labs. If not: download & install!

You can login to your VM using `student / HvA_student`.

Getting to know dig

On Linux, there are different tools to do DNS lookups, such as `host` and `nslookup`. For this lab, we'll be using the more extensive capabilities of `dig`.

Let's use `dig` to look up the IP-address of `www.hva.nl`:

```
dig www.hva.nl
```

You should see a **question section**, showing that we're looking for the A-record of `www.hva.nl` on the IN (Internet).

You should see an **answer section**, showing a few aliases for this domain name in the CNAME records and an IP address in the A record.

Finally, you should see the address of the local **DNS server** that performed the query for you.

1. What are aliases for `www.hva.nl`?
2. What is the IP address for `www.hva.nl`?
3. What is the IP address of the DNS server that performed the query?

- 1.
 - 2.
 - 3.

If you want to, you can also specify the DNS server you want to use for your queries. For this lab, you need to make sure that you're using a DNS server that supports DNSSec. One such server is Google's DNS server at `8.8.8.8`

Repeat your query, using Google's DNS server and verify that the server address has indeed changed:

```
dig @8.8.8.8 www.hva.nl
```

We can also use `dig` to find other types of DNS resource records. For example, to find the mail servers for the HvA domain, we need to query the MX record:

```
dig @8.8.8.8 hva.nl MX
```

4. What are the mail servers for hva.nl?

4.

One more thing we'll use later, is to use `dig` to query the DNS root domain. The root domain is indicated by a `.` (period). Let's find the root name servers by asking for the NS records:

```
dig @8.8.8.8 . NS
```

5. How many root name servers are there?

5.

Dig and DNSSec

The following table provides a list of some sites that do not have DNSSec implemented for their DNS records, some sites that do have DNSSec and finally some sites that have DNSSec set up with incorrect authentication.

Be aware that the status of some of these sites may have changed since this lab was written, so check your results accordingly!

Sites <i>without</i> DNSSec	Sites <i>with</i> DNSSec	Sites with <i>incorrect</i> DNSSec
www.nos.nl networksinthenews.com www.nu.nl	iana.org metebalci.com www.eurid.eu	dnssec-failed.org rhybar.cz

Query a few of these sites to see the different results. Pay attention to the first few lines of the response `dig` gives, just below the line `Got answer:`

- You'll see a number of flags. If DNSSec could verify the validity of the DNS records, it will show `AD` for Authenticated Data
- If the site does not use DNSSec to verify its records, the `AD` flag will not be shown
- If the site has DNSSec set up with false authentication information, the `AD` flag will not be shown either. You should also notice the `status: SERVFAIL` indicating a failure and the fact that no answer section has been provided, since the DNS records seem to be false.



Query the DNS records for sites without DNSSec, with DNSSec and with incorrect DNSSec. Make screenshots or pictures of your results to hand in.

If we insist on getting the DNS record for a site, even if DNSSec indicates the information cannot be trusted, we can run our query with the `CD` flag (checking disabled.):

```
dig @8.8.8.8 dnssec-failed.org +cd
```

6. What's the IP address for dnssec-failed.org?

6.

DNSSec Resource Records

To validate the authenticity of a set of DNS Resource Records, DNSSec defines four new record types:

- DNSKEY: keeps the public key to verify RRSIGs
- DS: keeps the digest of a DNSKEY RR
- RRSIG: keeps the digital signature of an RRset (set of Resource Records)
- NSEC: used for authenticated denial existence, meaning to show an RRset is not part of a signed zone



For one of the sites supporting DNSSec, use dig to request the DNSKEY records and the RRSIG records. Make screenshots or pictures of your results to hand in.

DNSSec verification

Verification using DNSSec works in a number of steps:

- We can trust a resource record to be correct, if it was signed with the private key of the authoritative name server. We can verify this, using its public key
- But can we trust the authoritative name server? We can, if it has been verified by the Top Level Domain Server (which we can check using its public key).
- To know if we can trust the TLD server, we'll want to check if it has been verified by a root server (which we can check using its public key)

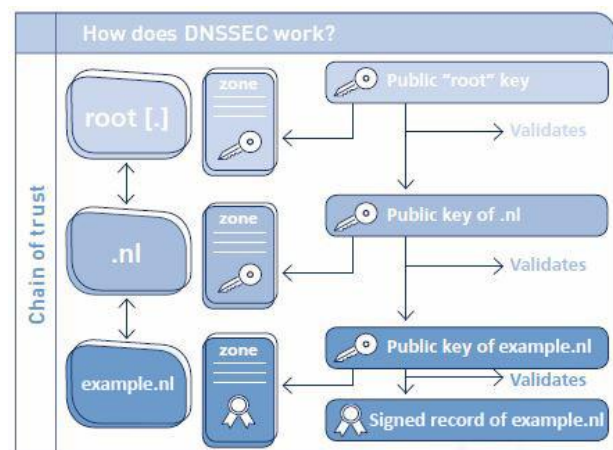


Figure 1

You'll see our chain of trust needs to start somewhere. IANA provides a trust anchor to validate the root keys at <https://www.iana.org/dnssec>. The root keys themselves are updated a few times a year during a ceremonial meeting of representatives.

To see the current root keys, enter:

```
dig @8.8.8.8 . DNSKEY
```

To be able to manually verify DNS records, we need to download the root keys to our own machine. Enter:

```
dig @8.8.8.8 . DNSKEY | grep -Ev '^($|;)' > root.keys
```

Then check if your download was successful:

```
cat root.keys
```

We can now use dig to verify the validity of a DNS record as follows. There'll be a lot of output, so we'll send it to `more` to view it page by page.

```
dig @8.8.8.8 +sigchase +trusted-key=root.keys <domain> | more
```



Try this check for one of the sites supporting DNSSec. Also try to validate a site that does not have DNSSec. Make screenshots or pictures of your results to hand in.



If we want to verify one of the sites that have invalid DNSSec authentication, make sure to run your query with the `+cd` flag, otherwise you will not receive an A-record to verify.



Try the check for one of the sites with invalid DNSSec authentication, using the `+cd` flag. Make screenshots or pictures of your results to hand in.

Complete the lab

To wrap up the lab, make a single document containing your answers and screenshots. Upload your document to DLO.

Sources and further reading

The following sources were used in creating this lab and provide a good starting point to learn more about DNS, DNSsec and dig.

- Waldner, 17-11-2010, DNSSEC verification with dig, retrieved on 1-11-2019 from <https://backreference.org/2010/11/17/dnssec-verification-with-dig/>
- Mete Balci, 2018, A short practical tutorial of dig, dns and dnssec, retrieved on 1-11-2019 from <https://metebalci.com/blog/a-short-practical-tutorial-of-dig-dns-and-dnssec/>
- IANA, DNSSEC Information, retrieved on 1-11-2019 from <https://www.iana.org/dnssec>

Figure 1: SIDN. DNSsec, retrieved on 30-10-2019 from <https://www.sidn.nl/veilig-internet/dnssec>